



UNIVERSAL COORDINATION PROTOCOL | UNICORD

Collateral Mobility Network

Execution Plan and Budget Proposal

A sovereign-grade cross-border compliance, messaging and settlement network for institutional asset mobility and asset-backed stablecoin issuance, built on the Universal Coordination Protocol.

Prepared for: The exclusive use by Solomon Cates and the Sovereign Technology group.

For: Sovereign wealth funds, pension plans, and ultra-high-net-worth treasuries.

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Executive Summary

What we deliver. Oko Labs delivers the Collateral Mobility Network: a single protocol on which an institutional asset moves, the compliance proof travels with it, and settlement is final, all at the line rate of the internet. It lets sovereigns, pension funds, and ultra-high-net-worth treasuries mobilize high-quality assets across borders privately, while producing cryptographic, replayable proof that every compliance check was fully executed and passed before value ever moved. The network is built on the Universal Coordination Protocol (UCP) and delivered as Unicord, with the Compliance Coordination product specification as the product guidepost.

Three problems, one fabric. Institutional capital faces three problems that the market today treats as separate. First, trillions in high-quality assets are functionally trapped inside legacy market infrastructure, forcing a costly trade-off between liquidity and yield. Second, compliance friction between jurisdictions kills cross-border transactions, traps capital in suspense, and exposes institutions to enforcement risk that now reaches board-level personal liability. Third, the messaging and settlement rails that would carry sovereign-grade value cannot, today, carry sovereign-grade trust at the scale and privacy these counterparties require. The Collateral Mobility Network resolves all three on one fabric: UCP coordination, compliance, and settlement are the same operation.

The product. The network realizes the Compliance Coordination specification (pre-validation with hold-and-release, dynamic policy interchange, jurisdiction-safe data handling, and on-chain regulatory reporting), and extends it to collateral mobility and to the issuance of an asset-backed stablecoin. An institution pledges high-quality assets into a bankruptcy-remote structure, the pledge and reserve coverage are continuously attested on-chain, and a stablecoin is minted against the pledged reserve. Every movement of that stablecoin, and every cross-border settlement it enables, carries an Execution Decision Object (EDO) that proves compliance was satisfied across all relevant jurisdictions before the transaction committed.

Privacy without disclosure. Privacy is achieved through data severability, not through zero-knowledge cryptography. Raw customer and asset data never leave their home jurisdiction. A receiving jurisdiction publishes its policies as a dynamically retrievable object, the originating side runs the receiver review function locally, and only a boolean verdict, a scoped decentralized identifier, and the EDO proof cross the boundary. The counterparty learns that the transaction is compliant without learning the underlying data.

The ask. We propose a twelve-month build-to-launch program that stands up the network, issues a live asset-backed stablecoin, proves the throughput and resilience required for global scale, and goes live with a closed group of counterparties across two to three jurisdictions under a full control framework. The committed program price is 19.66 million dollars, covering 47 full-time equivalents across the technology core and the business ecosystem at an average base of 216 thousand dollars per engineer. An initial 5 million dollar payment within fifteen days of signing funds rapid hiring; the remaining 14.66 million dollars is tied to eight execution-deliverable gates across the Execution Layer, Consensus engine versions, Stablecoin readiness, and Asset onboarding readiness, ending with a closed-group pilot live at month twelve. Oko Labs carries the engineering risk; payment after mobilization is milestone-based against

demonstrated results, and the figure is the gross program cost before any cost-sharing arrangement among participating counterparties.

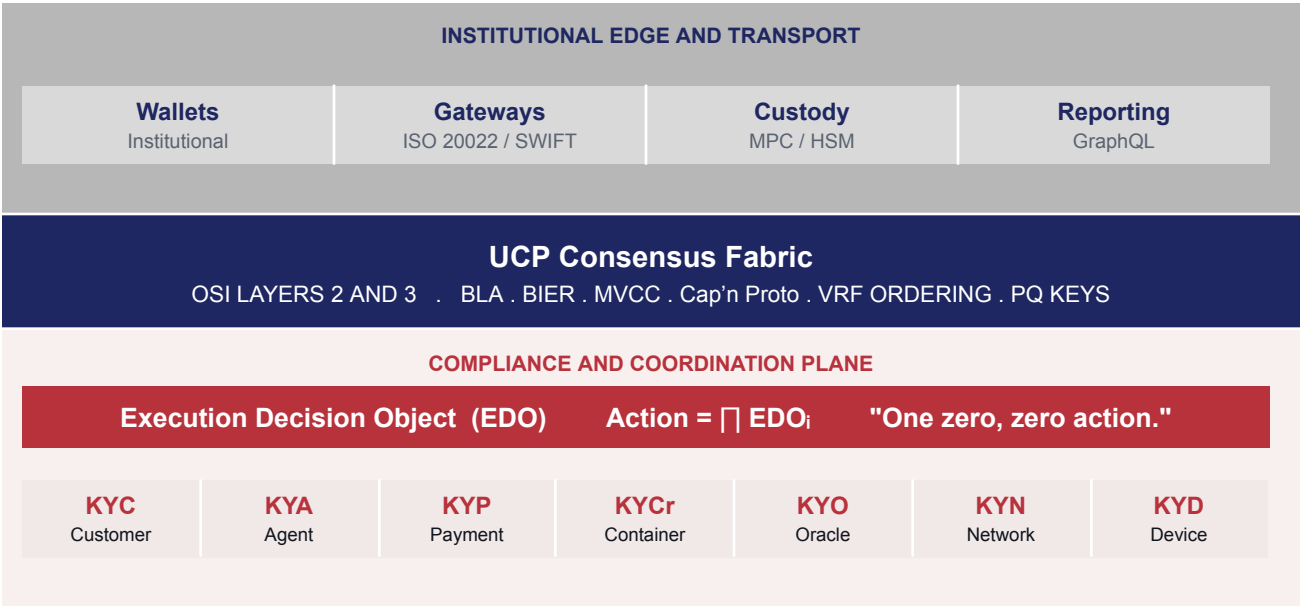


Figure 1. The Collateral Mobility Network: institutional edge and transport, the UCP consensus fabric at OSI Layers 2 and 3, and the compliance and coordination plane carrying the EDO and the seven KY-objects.

The Opportunity

2.1 Trapped collateral

Discovery work conducted with major Canadian and American pension plans and validated through interviews with sovereign and public-sector funds confirmed that institutional investors collectively hold trillions of dollars in high-quality assets that remain functionally trapped inside legacy market infrastructure. Four pain points surfaced consistently, and they are the demand signal for this network.

- **Intermediation costs erode yield.** Funds want to reduce dependence on broker-dealers, repo desks, and custodians. Even a ten basis point improvement in financing cost is material at multi-billion-dollar scale, and tokenized assets are viewed as a credible path to reducing spreads and counterparty risk.
- **Limited visibility into cash and positions.** Fragmented systems mean funds discover their true cash position too late in the day to capture the best short-term yields. Real-time, intraday reserve transparency is a recurring and unmet need.
- **Yield-curve mismatch and refinancing risk.** Many funds borrow short against long-duration assets, creating rollover risk and dependence on dealer balance sheets. An instrument aligned to long-duration liabilities materially reduces this risk.
- **Illiquidity limits portfolio agility.** Between forty and sixty percent of portfolios are illiquid. Funds want a mechanism to unlock liquidity without asset sales, enabling dynamic rebalancing and faster response to market dislocations.

Together these confirm demand for a controlled, institution-grade instrument that improves collateral mobility, reduces dependence on intermediaries, and lowers cost of capital. The mechanism the market keeps reaching for is a stablecoin backed by fund-owned assets, paired with the ability to move and settle it across jurisdictions without losing control of compliance.

2.2 Compliance friction kills cross-border flows

The way compliance is executed across the correspondent ecosystem today is structurally flawed. Checks are not run end-to-end before a payment is sent downstream. A transaction proceeds to the next intermediary without anyone knowing whether it should have been sent at all, and any failure along the chain imposes cost on every party and frustration on payer and payee. Sharing the results of compliance checks across participants is not available functionality today, and reporting those results to regulators is stitched together on demand rather than being systematically queryable.

Banks were never architected to manage data for regulatory purposes. A single customer can exist as six unlinked records across business lines, so when regulators ask whether the right checks were run and what the results were, institutions often cannot answer quickly or completely. The exposure attached to this gap is severe and now reaches the individual at the top of the institution.

Risk category	Stakeholder	Annual exposure, top-tier institution
Regulatory enforcement action	General Counsel, Board, CEO	630 million to 3 billion dollars (up to 4 percent of global turnover)
Failed-payment remediation	COO	Significant and growing
Suspense account and balance-sheet drag	CFO	Significant
Compliance remediation (people and systems)	COO, CCO	Growing annually, no control over trajectory
Personal liability under NIS2 (EU)	CEO, Board Chairman	Unlimited personal criminal exposure

This is the commercial lever. The Collateral Mobility Network turns clean, auditable, real-time compliance reporting into a first-class output of every transaction, and gives a board deterministic, replayable proof that every check was run.

2.3 Why existing rails and Layer-7 blockchains cannot serve these counterparties

Sovereign and institutional counterparties require an infrastructure that is private, provably compliant, resistant to surveillance and reordering, and able to carry settlement at scale for decades. Conventional Layer-7 blockchains (Ethereum and Ethereum-derived systems, and any architecture sharing their design) fail this bar for reasons that are architectural, not community-specific.

- **Mempool exposure and reordering (MEV).** On any chain where proposers see and order pending transactions, a well-positioned actor can front-run, sandwich, or censor them. Applied to settlement and provisioning, this is an operational weapon, not a financial nuisance. Over 600 million dollars has been extracted through these techniques on the most-documented chain alone.
- **Surveillance at the network edge.** Layer-7 peer-to-peer protocols use distinctive signatures that are trivially identifiable by deep packet inspection at any ISP or cable landing point, and all pending transactions are visible before execution. For a network carrying sovereign settlement, this hands an adversary passive intelligence as a built-in feature.
- **Shared global contract state.** On a shared state machine, a vulnerability in any contract can compromise others, and upgradeable or metamorphic contracts can pass an audit clean and turn malicious later. For long-horizon, state-sponsored adversaries this is a persistent backdoor.
- **No post-quantum path.** Pre-quantum keys will become vulnerable within the operating lifetime of any infrastructure built to last decades, and conventional chains have no viable migration path.

UCP eliminates these conditions by construction, as detailed in Section 5. There is no mempool, consensus runs at OSI Layers 2 and 3, secured via IPSec mesh (and OSI L2 network with bridges), and is indistinguishable from ordinary routing; execution is schema-bound rather than arbitrary, and identity keys are post-quantum from day one.

2.4 The customer

The network is designed for sovereign wealth funds, public and corporate pension plans, and ultra-high-net-worth treasuries, together with the custodians, settlement partners, and regulators that surround them. These counterparties share a non-negotiable set of requirements: assets must move globally and privately; every movement must carry provable, jurisdiction-specific compliance; data sovereignty must be enforced at the protocol layer; and the network must not be capturable by any single jurisdiction, vendor, or counterparty. The sections that follow describe how the product, the governance model, and the infrastructure meet each of these requirements.

The Product: Collateral Mobility on the Universal Coordination Protocol

The product is the Compliance Coordination specification realized on UCP and extended to collateral mobility and asset-backed stablecoin issuance. Every compliance component runs natively on-chain as a containerized workload within the network. There is no separate web infrastructure: the payment gateway, compliance engine, policy store, reporting layer, screening checks, and settlement logic are all containers deployed to jurisdictional nodes, with archive nodes providing persistence.

3.1 The core mechanism: the Execution Decision Object

The Execution Decision Object (EDO) is the atomic unit of the network. It is a tamper-evident, cryptographically signed compliance decision that travels with every transaction. An action is permitted only when the product of its constituent decision objects evaluates to a pass: $\text{Action} = \prod \text{EDO}_i$. The tagline captures the rule precisely: one zero, zero action. If any constituent decision evaluates to zero, the product is zero and the action does not occur. The EDO carries its own genealogy, so the reasoning that produced a compliance decision travels with the decision and can be replayed.

3.2 The KY-object model applied to collateral mobility

UCP expresses trust through seven composable KY-objects. Each entity in a transaction, whether human, machine, instrument, container, network path, or data source, carries a continuously attested, cryptographically verifiable trust profile. For collateral mobility, the pledged asset and its reserve coverage are expressed through the payment-instrument and oracle objects together with the container that hosts the pledge registry, so no new object type is required and the taxonomy stays clean.

Object	Entity	Role in the Collateral Mobility Network
KYC	Customer	The asset owner and counterparty institution. DID and verifiable-credential identity, revocable credential chain, decoupled from any single operator.
KYA	AI agent	Agents that read receiver policies and form-fit transactions to near-100 percent first-pass success. Model hash, authorization scope, and revocable delegation chain are attested.
KYP	Payment	The instrument and the movement. Instrument identity, ultimate beneficial ownership, amount and currency, economic substance, corridor risk, sanctions result, and settlement path. The pledged asset is carried here.
KYCr	Container	Every compliance workload, screening engine, policy store, and the pledge registry runs as an attested container (Bring Your Own Compliance), with SBOM, runtime posture, and data-severability scope sealed at instantiation.
KYO	Oracle	Reserve attestation, NAV, FX, and price feeds. Source methodology, freshness, and manipulation-resistance posture are attested, with full

Object	Entity	Role in the Collateral Mobility Network
		consumption lineage. This object proves reserve coverage above 100 percent.
KYN	Network	The path a transaction traverses, with beneficial ownership of each hop and a trust score that lets the network prefer trusted corridors.
KYD	Device	Custody hardware, HSMS, and signing devices. Continuous TPM 2.0 attestation, firmware integrity, and configuration baseline.

3.3 Pre-validation with hold-and-release

The core settlement mechanism is pre-validation with hold-and-release. Rather than sending a payment and letting it fail or stall downstream, the originating institution holds the transaction and balance in a reconciliation position, runs the full compliance callgraph against the receiver's dynamically published policies, and commits atomically only when both sides are satisfied. If every check passes, the payment is released for processing with a compliance reference for lookup. If any check fails, the balance is released back to the originator with a full audit trail of why. The transaction never enters the correspondent chain in a state that could fail.

3.4 Privacy through data severability

The network does not use zero-knowledge cryptography. Privacy is achieved through data severability: wallets and institutions choose what to reveal, and reveal it only on request. The originating institution performs a field projection so that the decentralized identifier it presents exposes only the data fields the receiver's policy actually requires. The originator can also define fields it will never disclose, which triggers a human-in-the-loop override or a pre-authorized mandate from the end client. The receiver review function executes inside the originating jurisdiction, so customer data never leaves its home jurisdiction; only a boolean verdict, a scoped DID, and the EDO proof cross the boundary.

3.5 Asset-backed stablecoin issuance

The network issues a stablecoin backed by fund-owned assets, giving institutions a controlled instrument to mobilize collateral without selling it. The lifecycle is pledge, register, mint, attest, settle, and redeem.

1. The asset owner pledges high-quality assets (for example short-dated government securities or whole-fund pledges) into a bankruptcy-remote special-purpose vehicle, perfected under the applicable secured-transactions framework. KYC and KYD attest the owner and the custody hardware.
2. The pledge is recorded in an on-chain registry, and reserve coverage is continuously attested by KYO oracles drawing on custody balances, NAV, and price feeds. Coverage is proven to exceed 100 percent.
3. The stablecoin is minted against the pledged reserve. Pre-mint verification confirms the asset is in custody and the reserve attestation is current.
4. Throughout its life the token moves and settles under hold-and-release, carrying an EDO at every step. A liquidity waterfall, conservative loan-to-value haircuts, redemption service levels, and swing pricing or gates provide a run-resistant design.

- On redemption the token is burned and the pledged asset is released, and reserve coverage is re-attested. Yield mechanics, whether accruing through the exchange rate (NAV per token) or a fixed rate, are configured per program and reported on-chain.

Token classification, the legal wrapper, and the accounting treatment are designed across the United States, Canada, the European Union, and the United Arab Emirates from the outset, so the instrument is regulator-aligned in every corridor it is intended to traverse.

3.6 Dynamic policy interchange and the compliance callgraph

Receiving institutions publish their compliance policies as versioned on-chain objects that the originating side queries at runtime. Policies can be updated at any time without rewriting commitments, so the network is a live runtime policy interchange rather than a static rulebook. The EDO genealogy is exposed through a GraphQL interface that treats every KY-object as a typed node with attested edges. A question such as is this transaction eligible for atomic settlement is answered not by a fixed decision tree but by traversing a subgraph that resolves the originating agent, the instrument's beneficial-ownership chain, the corridor trust score, the screening container's measurement, and the oracle's manipulation-resistance posture in one composable operation. New regulatory requirements become traversal conditions and edge predicates at the query layer, beneath the application layer and above the wire format, which is precisely where compliance belongs in a protocol that treats attestation as a network primitive.



Figure 2. The cross-jurisdiction compliance lifecycle. The receiver review function runs inside the originating jurisdiction; only a verdict, a scoped DID, and the EDO proof cross the boundary.

Jurisdiction, Policy and Governance

The network adheres to the UCP technical principles for jurisdiction definition, policy enforcement, and governance. These are the properties that let a sovereign or fiduciary join the network without ceding control, and they are the foundational answer to the question every operator asks before joining.

4.1 Jurisdiction definition

Each sovereign node operator configures its own compliance policy envelope across all asset classes simultaneously. A node operating under one central bank's policy enforces that policy for its financial flows; a node operating under a given securities or pensions regime enforces those rules for the instruments it touches. Each jurisdiction's node runs its own container stack, scoped to its data-sovereignty boundary, and data persisted to archive nodes is scoped to that jurisdiction. Cross-jurisdiction queries return only the boolean and DID outputs of the compliance callgraph, never raw customer or asset data. When a new nation joins, it does not integrate with Oko software; it adopts the protocol, runs its own validators, and enforces its own jurisdiction-specific rules, producing attestations interoperable with every other participant.

4.2 Policy enforcement at the protocol layer

Compliance policies are not bolted together at the application layer. They are expressions of the same policy engine, bound to the node at instantiation, attested through TPM-sealed measurement, and enforced at the same depth as routing and packet delivery. Access to data fields outside a container's declared severability scope is denied at the protocol layer, not the application layer, and logged in the EDO audit trail. Because execution is schema-bound through a zero-copy wire format, a container cannot emit a field that does not exist, read a field outside its scope, or return a forged attestation that conforms structurally while carrying attacker-controlled semantics. The compliance decision travels with the asset as a protocol-layer primitive that the asset cannot move without.

4.3 Governance: sovereignty and neutrality as complementary planes

The governance architecture has two interlocking planes. On the sovereignty plane, local policy over communications, asset movement, and settlement is absolute and technically tamper-evident: a node's compliance envelope is sealed at the protocol layer and is not negotiable at the application layer. On the neutrality plane, global protocol integrity is maintained through multi-jurisdictional Byzantine Lattice Agreement consensus over the primitives that define what compliant movement looks like: KY-object schema ratification, corridor risk parameters, new asset-class onboarding, and sanctions-list federation. Protocol upgrades require consensus across a weighted quorum of sovereign operators from multiple jurisdictions, where no single jurisdiction commands a unilateral majority and no actor, including the founding operators, can unilaterally alter the schemas. Together the two planes guarantee that the network cannot be captured by a foreign power at the governance layer, cannot be coerced at the node layer, and cannot be circumvented at the compliance layer.

4.4 Operator controls and update governance

Beyond the two-plane governance architecture, four operator controls are first-class properties of the network. They translate the requirement that no participant be forced into a state it has not chosen into specific, enforceable mechanisms at the protocol layer.

Asymmetric execution, symmetric consensus. Each participant runs the execution environment that suits its context, while every participant agrees on the same ledger of facts. Sovereign and institutional counterparties retain full control over their local stack without sacrificing global compatibility, and there is no privileged operator whose runtime defines the network's behavior.

Version-pinning with policy symmetry. A participant may choose to remain one or two versions behind the latest protocol release. Peers may, by policy, refuse to transact with non-current versions. Both are first-class operations on the network. No participant is forced to upgrade by network pressure alone, and any node carries a transparent, policy-readable record of its current version posture.

Privilegeless updates. No provider, including Oko Labs, can push an update to a participant's node without contracted consent. Update governance is held by the participant, not the vendor. This eliminates the supply-chain risk that vendor-controlled compromise (lawful or otherwise) introduces into long-horizon sovereign and institutional infrastructure.

Per-jurisdiction update authority. Each sovereign or institutional operator controls the schedule, scope, and acceptance criteria for protocol updates within its jurisdiction. The neutrality plane ratifies what compliant movement looks like; the sovereignty plane decides when a given version enters local production. The two operate independently by design.

Taken together, these properties make the long-term direction of Oko Labs, which is an open-source protocol and reference implementation released on a sequenced timeline, a continuation of the same operating posture rather than a discontinuity. Near-term, during the build-to-anchor period, the platform operates under controlled governance with the customer retaining configuration and policy authority. The open-source transition is intentional and sequenced.

Infrastructure for Throughput and Scale

Delivering a global cross-border messaging, compliance, and settlement network requires infrastructure that carries trust at line rate. UCP is not a blockchain riding on top of a network; it is a switching and routing-layer consensus fabric where each node is a virtual router with full forwarding and consensus capability, so consensus and network routing are the same operation.

5.1 Consensus at Layers 2 and 3

- **Byzantine Lattice Agreement (BLA)** provides sub-second deterministic finality with no mempool and no discretionary ordering, across a multicast domain designed for more than ten thousand nodes.
- **BIER multicast** propagates consensus as bit-indexed explicit replication at line rate, indistinguishable from ordinary IP routing, so there is no distinctive protocol signature to filter or surveil.
- **MVCC state with VRF-derived ordering** makes the transaction permutation deterministic and removes the proposer discretion that creates reordering attacks.
- **Cap'n Proto wire format** is a zero-copy, schema-enforced binary format, so field boundaries, types, and access permissions are enforced at the network layer before any container logic executes.

5.2 Throughput and finality targets

Because settlement state is derived from packet-layer consensus rather than queued in a rate-limited block cadence, throughput scales horizontally with the consensus domain rather than being capped by a global mempool. The program validates the following design targets.

Property	Target
Settlement finality	Sub-second, deterministic, no probabilistic confirmation wait
Pilot sustained throughput	Several thousand fully compliance-attested settlements per second per jurisdictional corridor, validated to one-million-endpoint scale
Scale model	Horizontal: additional corridors and validators raise aggregate throughput without re-architecting
Privacy overhead	None on the wire: data severability adds no proving step, unlike zero-knowledge approaches

5.3 Node topology

The network runs per-jurisdiction validator quorums with archive nodes providing persistence; the chain is the persistence layer, with no external database. For the twelve-month build-to-pilot we recommend a cloud-first validator topology for development and the closed-group pilot, with a lean dedicated cluster for

the jurisdictions that require it. Dedicated sovereign on-premise nodes are provisioned in the subsequent scale phase, when each sovereign stands up validators to its own specification. This sequencing is a deliberate cost-control decision, discussed in Section 8.

5.4 Resilience

View-aware routing gives every node chain-tip awareness, so a link failure triggers automatic path reconvergence and consensus continues within one view rotation, with no operator intervention and no failover runbook. Under partition or degraded conditions, Lifeboat Mode lets nodes on the minority side maintain local liveness and make trust decisions from cached KY-state, with post-quantum epoch checkpoints enabling resync after extended isolation. Self-healing and self-routing are inherent properties of a fabric where routing and consensus are unified, not features bolted onto a ledger.

5.5 Security posture

Identity keys are post-quantum (FIPS-206) with Ed25519 session keys, or potentially BLS signatures, on the CNSA 2.0 trajectory required for long-lived national-security and financial systems. The schema-bound execution model is a structural countermeasure to the upgradeable-contract and metamorphic-contract attacks that threaten shared-state chains: there is no upgrade key to steal and no implementation slot to overwrite, because changing what a compliance container does requires changing the schema it executes against, and schema changes are governed by consensus under the same sovereignty model as any other protocol state transition. The program includes FIPS 140-3 validation, two independent third-party security audits (protocol and product), and penetration testing before pilot go-live.

Audit on file. Oko Labs has completed an independent third-party audit. The full report is available for review on request under appropriate confidentiality terms.

Phased Execution: Twelve Months

The program runs in four overlapping phases over twelve months, with parallel teams executing concurrently. Phase A delivers the architecture foundation. Phase B (compliance core and stablecoin) and Phase C (consensus engine development and hardening) both begin in month 2, running in parallel with the late stages of Phase A. The consensus engine is the long pole, requiring at least eight months of solid heads-down engineering once the team is in seat. The Execution Layer therefore ships first, so that the Stablecoin and Asset-onboarding workstreams can build against the sandbox while the consensus engine is constructed underneath, with integration against real consensus starting at the v0.1 milestone (M6). Phase D launches the pilot in months 10 to 12. Each deliverable has a named owner and an exit gate; payment is tied to the gates in Section 9.

This is an aggressive schedule. The twelve-month window from team mobilization to closed-group pilot go-live across two to three jurisdictions, with a live asset-backed stablecoin, two clean third-party audits, FIPS 140-3 in progress, and independent reserve attestation, leaves no slack for late-binding scope. Holding the line requires three things: scope discipline against the pilot envelope, parallel execution across the protocol and product workstreams, and decisions taken at the gates rather than carried through them. Oko Labs has organized the program to meet that bar, and the milestone structure in Section 9 ties payment to the gates that measure it.

Months →	1	2	3	4	5	6	7	8	9	10	11	12
Phase A Foundation												
Phase B Compliance core + stablecoin												
Phase C Consensus engine + hardening												
Phase D Closed-group pilot + sustained												
Milestones and payouts	M1 \$5.00M		M2 \$1.80M		M3 \$1.80M	M4 \$1.80M	M5 \$1.50M	M6 \$2.00M	M7 \$1.80M		M8 \$2.00M	M9 \$1.96M

Figure 3. The twelve-month roadmap. Four overlapping phases (A through D) and nine milestone gates with their payouts: a 5 million dollar mobilization payment within 15 days of signing, followed by eight execution-deliverable gates ending with pilot readiness at month twelve.

Phase A. Foundation and Reference Architecture (Months 1 to 3)

Weeks	Deliverable	Owner and gate
1 to 2	Architecture decision brief; jurisdiction and corridor definition; product backlog from the Compliance Coordination spec	CTO and Product Lead. Architecture review
2 to 5	Reference architecture and non-functional requirements (security, resilience, observability, privacy); validator and archive topology; third-party audit team embedded in design reviews and sprint cycles from this point onward	Lead Architect and Audit Team. Design gate
3 to 6	EDO schema and KY-object schema design in Cap'n Proto; data-severability scope model	Compliance Container team. CTO review
4 to 8	Identity baseline: DID and VC schema, post-quantum key management plan (FIPS-206), Ed25519 session keys; FIPS 140-3 lab engaged	Identity and Crypto Engineer. FIPS initiated
6 to 10	Custody integration design (MPC and HSM connectors); legal and regulatory mapping across US, CA, EU, UAE begun	Custody Engineer and Compliance Architect. Architecture decision gate

Phase B. Compliance Core and Stablecoin Engine (Months 2 to 6)

Weeks	Deliverable	Owner and gate
5 to 12	Compliance Coordination core live on UCP sandbox: hold-and-release, dynamic policy store, jurisdictional containers, receiver-review-in-originator-jurisdiction; Stablecoin and Asset-onboarding teams begin building against the sandbox in parallel	Compliance Container team. Core milestone
7 to 15	KYP, KYC, KYA, KYCr live; sanctions, PEP, and Travel-Rule screening inside attested containers; ISO 20022 and SWIFT message gateway	Payments Engineer. CTO milestone
9 to 20	Stablecoin engine: pledge registry, pre-mint verification, mint and burn, reserve coverage above 100 percent attested by KYO	Stablecoin and Oracle Engineers. Stablecoin readiness gate
11 to 22	Liquidity waterfall, conservative LTV haircuts, redemption service levels, swing pricing and gates	Stablecoin Engineer. Controls review
13 to 24	GraphQL compliance callgraph and on-chain regulatory reporting; full transaction replay; asset onboarding readiness validated against the sandbox	Data and Reporting Engineer. Asset onboarding gate

Phase C. Consensus Engine Development, Throughput and Security Hardening (Months 2 to 11)

Weeks	Deliverable	Owner and gate
5 to 32	Consensus engine core development at OSI Layer 3: BLA over packet layer, BIER multicast, MVCC with VRF-derived ordering, Cap'n Proto wire format, post-quantum identity keys; first working consensus delivered at v0.1 (M6, month 8); v0.2 adding	Senior Protocol Engineers and Cryptography. v0.1 and v0.2 gates

Weeks	Deliverable	Owner and gate
	view-aware routing, Lifeboat Mode, and PQ epoch checkpoints (M7, month 9)	
24 to 32	Load and scale testing to one-million-endpoint scale; sub-second finality under load; resilience validation across partition, degraded-bandwidth, and satellite-only scenarios; benchmarks published	Load Testing Engineer and Security. Throughput and resilience gates
28 to 32	Consolidated third-party audit signoff at v0.2: protocol and product audit verdict issued from the continuous-audit cadence in flight since Phase A; targeted penetration testing on the consolidated codebase	CTO and Audit Team. Audit signoff
30 to 44	FIPS 140-3 crypto module progressing; any outstanding audit findings remediated in-sprint and re-verified through the same continuous cadence; controls and assurance mapping (SOC and COSO alignment); consensus engine v0.3 production-ready (M8, month 10 to 11)	Security Engineer and Compliance Architect. v0.3 hardening gate

Phase D. Closed-Group Pilot Go-Live and Sustained Operations (Months 9 to 12)

Weeks	Deliverable	Owner and gate
34 to 40	Pilot environment across two to three jurisdictions; counterparty onboarding and closed-group allowlists; operational playbooks and runbooks	Solutions and PMO. Onboarding gate
38 to 44	Pilot go-live: at least three counterparties, live but ring-fenced volume; stablecoin issued and redeemed end-to-end; cross-border settlement under hold-and-release	Lead Architect and PMO. Pilot go-live gate
42 to 48	Independent reserve attestation (Big-Four) and SOC-style controls evidence; regulator and auditor scoped read access validated	Compliance Architect and Auditor. Assurance gate
46 to 52	Sustained operations support; performance and uptime measured under sustained operation; scale-phase plan and handover	CTO and Engineering. Handover gate

Exit criteria

- Compliance Coordination core operational on UCP with hold-and-release, dynamic policy interchange, and data severability proven.
- Asset-backed stablecoin issued, moved, settled cross-border, and redeemed end-to-end, with reserve coverage above 100 percent continuously attested.
- Throughput validated to one-million-endpoint scale with sub-second finality, and resilience validated under at least four disruption scenarios.
- Two third-party security audits complete with no critical findings; FIPS 140-3 in progress; independent reserve attestation delivered.
- Closed-group pilot live across two to three jurisdictions with at least three counterparties and on-chain regulatory reporting.

Team and Headcount

Oko Labs stands up a 47-person team to deliver the program: approximately 33 engineers on the technology core (network engineering, execution environment, ledger engineering, systems and operations) and approximately 14 specialists on the business ecosystem layer (AI and ML, customer and vertical delivery, regional engagement). The average base salary across the team is 216 thousand dollars, with bonus structure bringing total cash compensation to 15.675 million dollars for the twelve months. The team is senior-weighted across all functions, reflecting the institutional and sovereign nature of the work.

7.1 Build team by function group

Function group	FTE	Primary scope
Founders and program leadership	2	CPO, CTO, CBO (fractional on this program), VP Engineering, PMO, technical writing, UX, administration
Network engineering	6	UCP consensus, BIER multicast, virtual routing, packet-layer protocol design
Execution environment	2	TEE and HSM, ADDC, container runtime (OCI, Docker, WASM)
Ledger engineering	11	Cryptography, SDK and wallet, smart contracts, decentralized identity, GraphQL archive, embedded database
Systems and operations	10	Kubernetes infrastructure, ops, support and maintenance, fullstack, junior security
AI and ML	4	KYA agent framework, vertical AI capability, customer SA
Customer and vertical delivery	11	Partner delivery; regional vertical and customer TPMs and SAs, including 5 Sovereigns.Technologies-dedicated seats (Section 7.2) and 5 DTR-aligned seats supporting the Kubernetes and container-runtime workstream that underlies KYCr
Build team total	~47	Twelve-month average across the program; peak headcount slightly higher accounting for ramp
Strategic advisors (not in FTE)	+2	Bradley Riss and Tom Zschach, senior advisory across protocol, identity, and institutional finance

7.2 Sovereigns.Technologies-dedicated engagement resources

Five of the seats listed under Customer and vertical delivery in Section 7.1 are specifically aligned to this engagement and its regional reach. These are not additions to the 47-person build team; they are dedicated allocations within it, placed alongside the engagement so that the customer has direct, regionally distributed engineering coverage for the lifetime of the program.

Role	Region	Function
Technical Program Manager, CBDC and Digital ID	UAE	Coordinates CBDC, exchange, and digital-identity workstreams; primary regional liaison for the program.
Solutions Architect, Vertical Team 1	APJC	Solution design for asset mobility and stablecoin onboarding across the Asia-Pacific corridor.
Solutions Architect, Vertical Team 2	UK / EU	Solution design and regulatory alignment for the UK and EU corridors, including MiCA and Travel-Rule integration.
Solutions Architect, Customer Team 1	Americas	Counterparty engagement and solution architecture for US and Canadian pension and sovereign fund onboarding.
Solutions Architect, CBDC and Digital ID	UAE	Solution architecture for CBDC, exchange, and digital-identity counterparties in the UAE and adjacent corridors.

Strategic partnerships and the DTR workstream. The five DTR-aligned seats inside Customer and vertical delivery are Oko Labs' contribution to the open-source Kubernetes and container-runtime workstream supported through Developer Technology Research (DTR), Oko's MoU partner for the containerized compliance execution environment that underlies KYCr. These seats are part of the 47-person build team and are covered by the program budget in Section 8; the DTR partnership itself, together with the other MoU partners (FYEO, Mindex, GMEX/Zero13, Minera, SpaceArmor), extends the network's reach without requiring additional dedicated headcount from the customer.

7.3 Leadership and advisory

The program is led by Anoop Nannra (CPO), a serial founder in digital assets and blockchain infrastructure, and Jeff Venable (CTO), a distinguished engineer in security, blockchain, networking, and distributed protocol architecture with more than forty patents. Ashley Trick (CBO) leads commercial structuring. The advisory bench includes Tom Zschach, former CIO of SWIFT, alongside senior practitioners from institutional finance and digital-asset custody. This blend of protocol depth, institutional-finance experience, and enterprise delivery is the basis for carrying the engineering risk on a milestone-based program.

Budget

The budget is built bottoms-up from the Oko Labs hiring plan and operations budget for the twelve-month build. Personnel reflects 47 full-time equivalents, with the loaded rate structure bringing total cash compensation to 15.675 million dollars. Operations and capex cover facility, equipment, external services, travel, and tooling, and total 3.779 million dollars. The program total of 19.45 million dollars is the gross figure before any cost-sharing arrangement among participating counterparties.

8.1 Cost by category

Category	Detail	Cost
Personnel	47 FTE; fully loaded labor rate costs reaching 15.675 million dollars across the twelve months	\$15,675,000
External services	Legal counsel and structuring (250k); HR and recruiting (500k); third-party security audit, FYEO (500k); specialized engagement services (702k)	\$1,952,000
Travel and events	Customer travel (300k); quarterly offsites (300k); program kickoff (100k)	\$700,000
Facility and office	Office and lab space (595k); productivity software and tools, including Claude enterprise, GitHub, and team collaboration subscriptions (282k)	\$877,725
Equipment	Edge node hardware kit, one development and one static node per team member (100k); team laptops (150k)	\$250,000
Total program cost	Bottoms-up from the hiring plan and operations budget; gross figure before cost-sharing	\$19,454,725

8.2 Build composition and cost-share posture

The revised figure reflects what is required to deliver a production-grade settlement network at sovereign and institutional scale: the technology core plus the business ecosystem needed to onboard counterparties, regulators, and partners in production. Three observations frame what this budget covers and how it can be borne.

First, the technology core is roughly two-thirds of the build. Network engineering, execution environment, ledger engineering, and systems and operations together (33 FTE) deliver UCP consensus, the compliance container framework, the stablecoin engine, throughput and resilience validation, and the post-quantum hardening described in Sections III through VI. The remaining 14 FTE constitute the ecosystem layer: AI and ML for the KYA agent framework, customer and vertical delivery (regional TPMs and SAs), and the partnership-aligned seats (Sovereigns.Technologies-dedicated and DTR-aligned). Both layers are required for a network that institutional counterparties can actually onboard to in twelve months.

Second, regional ecosystem investment is what makes the pilot real. Five Sovereigns.Technologies-dedicated seats (UAE TPM, regional SAs in APJC, UK/EU, Americas, and a second UAE SA) plus five DTR-aligned seats supporting the Kubernetes and container-runtime workstream that underlies KYCr together represent the on-the-ground capacity for counterparty onboarding and operational continuity. Without these, a closed-group pilot across two to three jurisdictions in twelve months is not credible.

Third, cost-sharing is a material option. The 19.45 million dollar figure is the gross program cost. In comparable institutional engagements, program cost has been shared across participating pilot counterparties; the strategic investment from Oko Labs technology partners further reduces the net commitment to any single sponsor. The cost-sharing structure, the proportions, and the participating counterparties are open commercial questions that will be resolved alongside the master services agreement.

Net conclusion. The committed program cost is 19.45 million dollars covering 47 FTE across the technology core and the business ecosystem over twelve months. The figures above are bottoms-up from the hiring plan and operations budget; no implicit indirect or contingency markups are added. Material scope changes (additional jurisdictions, on-premise sovereign nodes in the build phase, or a larger pilot counterparty set) are handled through change control rather than absorbed silently into the line items.

Milestone-Based Payment Structure

Oko Labs carries the engineering risk; the customer pays on demonstrated results. Milestone payments are tied to execution deliverables across the Execution Layer, the Consensus engine, Stablecoin readiness, and Asset onboarding workstreams. The first payment, due within fifteen days of signing, funds the rapid hiring required to mobilize the 47-person build team. The eight subsequent payments are tied to verifiable readiness gates and sum, with M1, to the program cost of 19.45 million dollars. Two things drive the sequencing. First, the consensus engine requires at least eight months of solid heads-down engineering once the team is in seat, which puts production-grade consensus (v0.3) at month 10 to 11, and pilot integration at month 12. Second, the Execution Layer ships first so that the Stablecoin and Asset-onboarding teams can build and prototype against the sandbox while the consensus engine is being constructed underneath. Consensus is delivered at OSI Layer 3 over IP routing. The architecture is also L2-capable, but operating consensus at OSI Layer 2 requires direct relationships with internet backbone providers (Tier-1 carriers and exchange operators) and corresponding GTM enablement; that workstream is not in scope for this program and would be a follow-on engagement, contingent on the sponsor network convening those providers. The third-party audit team is embedded in development sprints from Phase A onward; the audit signoffs visible in the M8 milestone reflect the verdict from that continuous cadence rather than a discrete end-of-program audit event. Change orders for material scope changes are processed separately and do not draw from these milestones.

Milestone	Payment	Trigger	Timing
M1	\$5,000,000	Signed statement of work; payment delivered within 15 days of signing to fund hiring across the 47-person build team; team mobilization begun	Within 15 days
M2	\$1,800,000	Execution Layer v0.1: developer sandbox live; container runtime (KYCr foundation) operational; first attestation framework; Stablecoin and Asset-onboarding teams can begin building against the sandbox	Month 3
M3	\$1,800,000	Execution Layer v0.2: full KYCr framework with SBOM and runtime attestation; data-severability scope model; compliance container deployment; receiver-review-in-originator-jurisdiction operational	Month 5
M4	\$1,800,000	Stablecoin readiness: pledge registry, pre-mint verification, mint and burn, reserve coverage attestation by KYO above 100 percent, liquidity waterfall and redemption controls, all operating against the sandbox	Month 6
M5	\$1,500,000	Asset onboarding readiness: KYP, KYC, KYA live; sanctions, PEP, and Travel-Rule screening inside attested containers; ISO 20022 and SWIFT message gateway operational against the sandbox	Month 7

Milestone	Payment	Trigger	Timing
M6	\$2,000,000	Consensus engine v0.1 at OSI Layer 3: BLA consensus over IP routing, MVCC with VRF ordering, no mempool, deterministic finality benchmark established; Execution Layer applications begin integration testing against real consensus	Month 8
M7	\$1,800,000	Consensus engine v0.2: view-aware routing, automatic path reconvergence, Lifeboat Mode under partition, post-quantum epoch checkpoints; resilience validated under degraded conditions	Month 9
M8	\$2,000,000	Consensus engine v0.3 (production-ready) at OSI Layer 3: throughput validated to one-million-endpoint scale, sub-second finality under load, third-party audit signoff issued at v0.3 from the continuous-audit cadence running since Phase A, FIPS 140-3 in progress (OSI Layer 2 enablement is contingent on backbone provider partnerships and is not in scope for this milestone)	Month 10 to 11
M9	\$1,754,725	Pilot ready: closed-group pilot live across two to three jurisdictions, at least three counterparties, stablecoin issued and redeemed end-to-end, cross-border settlement under hold-and-release, independent reserve attestation, sustained operations handover	Month 12
Total	\$19,454,725	Gross program cost before any cost-sharing arrangement among participating counterparties	

Risk Register

Risk	Lk	Im	Mitigation
Cross-jurisdiction legal and token classification exceeds allowance	M	H	Engage counsel in all four regimes in Phase A; sequence the stablecoin to the jurisdictions cleared first; legal overruns escalated through the change-order process
FIPS 140-3 validation exceeds twelve months	M	H	Engage the lab immediately; use FIPS-validated third-party crypto as interim; full certification required before production
Critical security finding emerges late in the program	L	M	Third-party audit team is embedded in development sprints from Phase A and participates in design reviews, code reviews, and architecture decisions; findings surface and are remediated in-sprint at small magnitude rather than batched for end-of-program audit; the audit team's cadence is matched to engineering velocity by contract
Throughput bottleneck at scale	M	M	Begin load testing in Phase B; identify bottlenecks progressively at 100k, 500k, and 1M endpoints; horizontal corridor scaling
Reserve attestation and custody integration delays	M	M	Custody design in Phase A; start with one custodian and one KYC and sanctions provider; KYO oracle framework decoupled from any single feed
Scope expansion toward global production within twelve months	M	H	Hold pilot scope; sovereign on-premise nodes and additional corridors moved to the scale phase; scope changes processed through change control rather than absorbed into the program

Next Steps and Defined Terms

#	Action	Owner
1	Review this proposal and confirm pilot scope, target jurisdictions, and initial counterparties	Sponsor and Oko Labs
2	Technical deep-dive: CTO presents the security and scale assessment with a live compliance and hold-and-release demonstration	Oko CTO and Sponsor
3	Confirm the stablecoin program: asset classes for pledge, custody, and the legal wrapper across target jurisdictions	Sponsor, Oko, Counsel
4	Statement of work execution and team mobilization (M1)	All parties
5	Phase A architecture decision and reference architecture (M2)	Oko Engineering

Defined terms

UCP, the Universal Coordination Protocol. Unicord, the product implementation of UCP. EDO, the Execution Decision Object, the signed compliance decision that travels with every transaction, governed by the rule Action equals the product of its constituent EDOs. KY-objects, the seven composable trust objects (Customer, Agent, Payment, Container, Oracle, Network, Device). BLA, Byzantine Lattice Agreement. BIER, bit-indexed explicit replication multicast. BYOC, Bring Your Own Compliance. Data severability, the privacy model in which only required fields are disclosed on request, used in place of zero-knowledge proofs.

Discovery Questionnaire

This questionnaire gathers the information required to finalize scope, pricing, and timeline for the twelve-month build and the phases beyond it. The scope, jurisdiction, and commercial sections most directly shape the engagement structure. Responses can be returned alongside the signed statement of work or in advance of the technical deep-dive listed in Section 11.

A.1 Project scope and definition of done

These answers most directly shape the engagement structure, price, and timeline.

- Define precisely what the solution layer on top of the platform must deliver: the applications, workflows, user roles, interfaces, and reporting outputs that constitute an accepted deliverable.
- Which asset and collateral types are in scope (for example government securities, money-market positions, fund units, equities), and what are their custody and valuation characteristics?
- Is asset-backed stablecoin issuance required within the initial twelve-month engagement, or is it a later phase?
- What settlement model is required (delivery-versus-payment, payment-versus-payment, atomic settlement), and which asset legs are involved?
- What reporting, attestation, audit, and replay outputs are required, in what format, and to whom must they be delivered?
- Will counterparty teams or external developers want to build against the Execution Layer sandbox during the program? If so, from which counterparties, and at what scale?
- What does the minimum acceptable initial delivery look like, as distinct from the full vision?

A.2 Throughput, scale, and performance targets

Concrete numbers separate a pilot-scale build from a global-production build.

- What are the target settlement volume and value for the pilot, and the production figures envisioned thereafter?
- How many endpoints and counterparties must the pilot support, versus production? The program validates throughput at one-million-endpoint scale; sponsor-specific targets above or below that should be flagged.
- What are the latency and settlement-finality expectations?
- What availability, disaster-recovery, and business-continuity requirements apply?

A.3 Jurisdictions and regulatory

Cross-jurisdiction legal and classification work is the largest single driver of cost and schedule risk.

- Which specific jurisdictions are in scope for the pilot, both originating and receiving?

- Which entity holds the relevant licenses and regulatory relationships in each jurisdiction?
- Is the counterparty a regulated entity able to host or operate a compliance execution environment?
- Do token-classification and supporting legal opinions already exist for each jurisdiction, or does that analysis fall to the engagement?
- Which regulatory regimes and reporting obligations apply (for example Travel Rule, MiCA, and local equivalents)?

A.4 Counterparties, custody, and integration

Integration surface area and custody posture determine engineering scope.

- Who are the closed-group counterparties for the pilot, and how many?
- Is there an existing custody arrangement, or is new MPC and HSM custody integration required?
- Which counterparty systems must be integrated (treasury, position-keeping, messaging such as SWIFT or ISO 20022)?
- Which identity, KYC, and sanctions-screening providers are already in place?
- What oracle and market-data sources are required for valuation, NAV, and FX?

A.5 Commercial framework and engagement structure

These answers determine how the engagement is funded, contracted, and governed.

- What is the budget framework and funding model for the engagement, given the 19.66 million dollar program cost over twelve months?
- Does the funding model accommodate the 5 million dollar mobilization payment within 15 days of signing, with the remaining 14.66 million tied to the eight execution-deliverable gates described in Section 9?
- Is the sponsor open to a cost-sharing structure across two or more participating pilot counterparties, or is single-sponsor funding preferred?
- Are there counterparties beyond the initial pilot group that the sponsor would convene as future scale-phase participants?
- Is there appetite for a strategic relationship beyond a standard vendor engagement?
- Are there exclusivity expectations, and to which vertical or jurisdiction would they apply?
- Is the counterparty willing to act as a named reference or joint case study?

A.6 Success criteria and continuation

A shared definition of success de-risks both the pilot and the path to production.

- How does the counterparty define a successful pilot, and what specific criteria would trigger a production phase?
- What is the envisioned scope, scale, and timing of the production phase?
- Who are the decision-makers and what are the criteria for continuation?

A.7 Timeline, governance, and procurement

These answers confirm whether the twelve-month target is firm and how the work will be approved.

- What is the required start date, and are there fixed external deadlines driving the twelve-month target?
- What is the procurement path, including approval stages and the authorized signer?
- What governance and steering structure is expected during delivery?
- Does the sponsor have a preferred third-party audit firm for the continuous-audit cadence described in Section 9, or is selection from Oko's preferred panel (including FYEO, the named audit partner in the budget) acceptable?
- What security review, due-diligence, or audit requirements must the engagement satisfy?

A.8 Network operations and follow-on engagements

Certain capabilities are out of scope for the twelve-month program and would be executed through follow-on engagements. These answers shape what gets queued.

- OSI Layer 2 enablement requires backbone provider partnerships and is not in scope for this program (Section 9). Does the sponsor have existing relationships with Tier-1 carriers or internet exchange operators that could support an L2 enablement workstream as a follow-on engagement, and what is the realistic timeline for bringing those parties to the table?
- Are there strategic technology partners the sponsor wishes to bring into the network, beyond Oko's existing MoU partners (including FYEO, Mindex, GMEX/Zero13, Minera, SpaceArmor, and DTR)?
- What is the sponsor's view on the long-term governance posture: sovereign-operator quorum, multi-jurisdictional Byzantine consensus on schema ratification, or other structures (Section 4)?
- Is the sponsor interested in early node-operator participation (running per-jurisdiction validator quorums), or solely as a counterparty consuming the network?

Sovereign Network Instantiation on Oko

A sovereign network on Oko is defined as configuration, not as a bespoke build. The properties a sovereign operator cares about, the validator set, the jurisdictional execution and archival boundaries, routing policy, compliance and disclosure policy, and the upgrade path, are expressed as declarative objects under consensus rather than encoded into a one-off deployment. This is what allows Sovereign Technologies to stand up and operate its own network posture without commissioning custom protocol work, and it is the protocol-level mechanism behind the two-plane governance model in Section 4.

B.1 Declarative network objects

The reflection layer treats network configuration as first-class, versioned objects, analogous to Kubernetes Custom Resource Definitions. Validator leases, execution-environment container versions, routing policies, compliance and disclosure policies, and upgrade manifests are each represented as objects in the ledger. Because these objects are part of consensus state, every node observes the same configuration and applies changes atomically. Configuration changes are themselves validated through consensus and leave a durable, auditable record of every governance action, which is the same property the per-jurisdiction update authority in Section 4.4 depends on.

B.2 Authoring and audit surface

Operators do not manipulate the canonical wire format directly. A codec framework provides deterministic, schema-validated mappings between canonical ledger objects and external representations, including JSON, YAML, and Kubernetes CRD descriptors. A sovereign operator authors and inspects its declarative network objects through standard JSON and YAML tooling, and a Kubernetes operator can treat a ledger state object as a CRD directly. The result is an infrastructure-as-code workflow over ledger state: the network posture is version-controlled, reviewable, and diffable with tooling the operator's engineers already run, rather than through a proprietary console.

B.3 Canonical format

The native, consensus-canonical format is Cap'n Proto. JSON and YAML are first-class codec-rendered views over that canonical form, not a parallel native format. This distinction is the source of the capability's strength: configuration authored as JSON is bound to a canonical, consensus-validated, versioned object rather than ingested as free-form text. The reflection layer keeps every object self-describing, so the JSON view is always a faithful, typed projection of committed state, and the same object can be rendered to the format an auditor or downstream system expects without affecting consensus or execution semantics.

B.4 Relevance to Sovereign Technologies

Three properties follow directly from this model and map to requirements raised in discovery. First, version-pinning and update authority: because execution-environment versions and upgrade manifests are declarative objects under the operator's control, a sovereign node can decline a forced upgrade, with

peers enforcing version compatibility through policy rather than coerced rollout. This is the protocol-level basis for the operator controls described in Section 4.4. Second, jurisdictional containment: execution runs inside geo-fenced containers bound to the operator's jurisdiction, with receipts retained in jurisdictional archives that never cross the boundary, while only compact proofs reach global consensus. Third, atomic and auditable change management: updates and policy changes roll out coherently through the same mechanism used for containerized contract execution, with a complete record in the reflection layer available for regulatory review.

OKO LABS

Coordination belongs in the network fabric.

Not in the application layer. Not in correspondent chains.

Not in policy assembled after the fact.

Okolabs moves coordination to where packets, policy, and proof move together.

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For inquiries: ir@okolabs.xyz